ARFKEN

López Merino Mauricio ⟨Bra | Ket⟩

⟨Bra | Ket⟩ ARCHIVE

INFINITE SERIES

PhD Candidate Preparation - University of Cambridge

Arfken Problems Chapter one

Problem 1.1.1 a) Prove that if $lim_{n\to\infty} n^p u_n = A < \infty, p > 1$, the series $\sum_{n=1}^{\infty} u_n$ converse.

Solution :

For Kummer's theorem we know if this is true the series converse :

$$\lim_{n\to\infty}\left(a_n \frac{u_n}{u_{n+1}} - a_{n+1}\right) \geq C > 0,$$

In this case we have :

$$lim_{n\to\infty} n^p u_n = A \Longrightarrow As\ n \to \infty, u_n \sim \frac{A}{n^p},$$

For $a_n = n$:

$$\lim_{n\to\infty}\left(a_n \frac{u_n}{u_{n+1}} - a_{n+1}\right) = \lim_{n\to\infty}\left(n \frac{\frac{A}{n^p}}{\frac{A}{(n+1)^p}} - (n+1)\right),$$

$$\Longrightarrow \lim_{n\to\infty}\left(n \frac{A(n+1)^p}{An^p} - (n+1)\right) = \lim_{n\to\infty}\left(n \frac{(n+1)^p}{n^p} - (n+1)\right),$$

$$\Longrightarrow \lim_{n\to\infty}\left(n\left(1+\frac{1}{n}\right)^p - n - 1\right),$$

For $n \to \infty$ $\left(1+\frac{1}{n}\right)^p \approx 1 + \frac{p}{n}$ binomial newton approximation :

$$\Longrightarrow \lim_{n\to\infty}\left(n\left(1+\frac{p}{n}\right) - n - 1\right) = \lim_{n\to\infty}(n - n + p - 1) = \lim_{n\to\infty}(p-1) = p - 1,$$

Remember $p > 1 \Longrightarrow \{\forall p \exists C |\ C > 0\ \wedge p - 1 = C\}$ for them :

$$\therefore \lim_{n\to\infty}\left(a_n \frac{u_n}{u_{n+1}} - a_{n+1}\right) \geq C > 0.$$

By Kummer's theorem, since $C = p - 1 > 0$ for $p > 1$ the series $\sum_{n=1}^{\infty} u_n$ convergent.

b) Prove that if $lim_{n\to\infty}\, nu_n = A > 0$, the series diverges. (The test fails for $A = 0$.)

Solution :

For Kummer's theorem we know if this is true the series diverges :

$$\lim_{n\to\infty}\left(a_n\frac{u_n}{u_{n+1}} - a_{n+1}\right) \leq 0,$$

In this case we have :

$$lim_{n\to\infty}\, nu_n = A \Longrightarrow As\; n\to\infty, u_n \sim \frac{A}{n},$$

For $a_n = n \;\wedge A \neq 0$:

$$\lim_{n\to\infty}\left(a_n\frac{u_n}{u_{n+1}} - a_{n+1}\right) = \lim_{n\to\infty}\left(n\frac{\frac{A}{n}}{\frac{A}{(n+1)}} - (n+1)\right),$$

We can see here if $A = 0$ the test fails for $u_n = \frac{A}{n} = 0$ this trivial convergent at 0 :

$$\Longrightarrow \lim_{n\to\infty}\left(n\frac{A(n+1)}{An} - (n+1)\right) = \lim_{n\to\infty}\left(n\left(1+\frac{1}{n}\right) - (n+1)\right),$$

$$\Longrightarrow \lim_{n\to\infty}\big((n+1) - (n+1)\big) = 0,$$

$$\therefore \lim_{n\to\infty}\left(a_n\frac{u_n}{u_{n+1}} - a_{n+1}\right) \leq 0.$$

These two tests, known as **limit tests**, are often convenient for establishing the convergence of a series. They may be treated as comparison tests, comparing with

$$\sum_n n^{-q}\,, 1 \leq q < p.$$

In conclusion, these results allow us to establish the convergence or divergence of a series by comparing its asymptotic behavior with the p-series $\sum \mathrm{n}^{-\mathrm{q}}$, using it as a reference scale.

1.1.2 If $\lim_{n\to\infty} \frac{b_n}{a_n} = K$, constant with $0 < K < \infty$, show that $\sum_n b_n$ converges or diverges with $\sum a_n$.

Solution :

Remember the lim define:

$$\lim_{n\to\infty} \frac{b_n}{a_n} = K,$$

For any $\epsilon > 0$, there exists an integer N such that for all $n \geq N$:

$$K - \epsilon < \frac{b_n}{a_n} < K + \epsilon,$$

For since $a_n > 0$, we can multiply the inequality:

$$(K - \epsilon)a_n < b_n < (K + \epsilon)a_n,$$

For other side :

$$\frac{b_n}{K - \epsilon} < a_n < \frac{b_n}{K + \epsilon},$$

By the comparison test:

1. If $\sum a_n$ converges, then $\sum(K + \epsilon)a_n$ converges , so $\sum b_n$ converges, conversely If $\sum b_n$ converges, then $\sum \frac{b_n}{K-\epsilon}$ converges , so $\sum a_n$ converges
2. If $\sum a_n$ diverges, then $\sum(K - \epsilon)a_n$ diverges, so $\sum b_n$ diverges, conversely If $\sum b_n$ diverges, then $\sum \frac{b_n}{K+\epsilon}$ diverges, so $\sum a_n$ diverges.

Thus, $\sum b_n$ converges or diverges with $\sum a_n$.

1.1.3 (a) Show that series $\sum_{n=2}^{\infty}\frac{1}{n(\ln n)^2}$ converges.

Solution :

Consider the function $f(x)=\frac{1}{x(\ln x)^2}$.

1. Conditions: For $2<x$, $f(x)$ is continuous, positive ,and monotonal decreasing.
2. Integral Test: The series converges if and only if the following improper integral converges:

$$I=\int_{2}^{\infty}\frac{dx}{x(\ln x)^2},$$

Evaluation :

Let $u=\ln x$, then $du=\frac{dx}{x}$.

When $x=2, u=\ln 2$. When $x\to\infty, u\to\infty$.

$$I=\int_{\ln 2}^{\infty}\frac{du}{u^2}=\left[-\frac{1}{u}\right]_{\ln 2}^{\infty},$$

$$I=\lim_{b\to\infty}\left(-\frac{1}{b}+\frac{1}{\ln 2}\right)=0+\frac{1}{\ln 2}=\frac{1}{\ln 2},$$

Conclusion:

Since the integral converges to $\frac{1}{\ln 2}$, the series converges.

(b) By direct addition $\sum_{n=2}^{100,000}\frac{1}{n(\ln n)^2}=2.02288$, use the next equation :

$$\int_{N+1}^{\infty}f(x)dx\le\sum_{n=N+1}^{\infty}a_n\le\int_{N+1}^{\infty}f(x)dx+a_{N+1}\,.$$

To make a five significative figure of the sum of this series.

Solution:

We have $N=100,000$ $f(x)=\frac{1}{x(\ln x)^2}$ for them :

$$\int_{N+1}^{\infty}f(x)dx=\int_{100,001}^{\infty}\frac{dx}{x(\ln x)^2},$$

Evaluation :

Let $u = \ln x$, then $du = \frac{dx}{x}$.

When $x = 100{,}001, u = \ln 100{,}001$. When $x \to \infty, u \to \infty$.

$$I = \int_{\ln 100{,}001}^{\infty} \frac{du}{u^2} = \left[-\frac{1}{u}\right]_{\ln 100{,}001}^{\infty},$$

$$I = \lim_{b\to\infty}\left(-\frac{1}{b} + \frac{1}{\ln 100{,}001}\right) = 0 + \frac{1}{\ln 100{,}001} = \frac{1}{\ln 100{,}001} \approx 0.08686,$$

And for other side :

$$\int_{N+1}^{\infty} f(x)dx + a_{N+1} = \int_{100{,}001}^{\infty} \frac{dx}{x(\ln x)^2} + \frac{1}{100{,}001(\ln(100{,}001))^2},$$

$$\int_{N+1}^{\infty} f(x)dx + a_{N+1} \approx 0.08686 + \frac{1}{100{,}001(\ln(100{,}001))^2},$$

For them :

$$\int_{N+1}^{\infty} f(x)dx \le S_N \le \int_{N+1}^{\infty} f(x)dx + a_{N+1}.$$

$$2.10974 \le S_\infty \le 2.10974,$$

Conclusion: Since both bounds agree to the first five significant figures, the sum of the series is **2.1097**.

1.1.4 Gauss' test is often given in the form of a test of the ratio :

$$\frac{u_n}{u_{n+1}} = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0}.$$

For what values of the parameters a_1 and b_1 is there convergence? divergence?

Solution :

Remember de Gauss' test:

$$\frac{u_n}{u_{n+1}} = 1 + \frac{h}{n} + \frac{B(n)}{n^2},$$

Where $B(n)$ is bounded function ,for then series is:

Converge if $h > 1$, diverge if $h \leq 1$.

Now :

$$\frac{u_n}{u_{n+1}} = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0} * 1 = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0} \frac{\frac{1}{n^2}}{\frac{1}{n^2}} = \frac{1 + \frac{a_1}{n} + \frac{a_0}{n^2}}{1 + \frac{b_1}{n} + \frac{b_0}{n^2}},$$

We use the Taylor expansion $(1 + x)^{-1} = 1 - x + x^2 - \cdots$;

$$\frac{u_n}{u_{n+1}} = \left[1 + \frac{a_1}{n} + \frac{a_0}{n^2}\right]\left[1 - \left(\frac{b_1}{n} + \frac{b_0}{n^2}\right) + \cdots\right],$$

$$\frac{u_n}{u_{n+1}} = 1 + \frac{a_1}{n} + \frac{a_0}{n^2} - \left(\frac{b_1}{n} + \frac{b_0}{n^2}\right) - \frac{b_1 a_1}{n^2} + \cdots,$$

$$\frac{u_n}{u_{n+1}} = 1 + \frac{1}{n}(a_1 - b_1) + \frac{1}{n^2}(a_0 - b_0 - b_1 a_1 + b_1^2) + \cdots,$$

Is easy now identify $h = a_1 - b_1$ and $B(n) = a_0 - b_0 - b_1 a_1 + b_1^2 + O\left(\frac{1}{n}\right)$ where is a bounded function, for the series is Converges if $a_1 - b_1 > 1$, diverges if $a_1 - b_1 \leq 1$.

1.1.5 Test for convergence:

$$(a)\sum_{n=2}^{\infty}\frac{1}{\ln n},$$

Solution :

We use Cauchy Condensation test :

Let $\{a_n\}$ be a sequence of non-negative real numbers that is monotonically decreasing If $(a_n \geq a_{n+1} \geq 0)$. The infinite series $\sum_n^{\infty} a_n$ converges if and only if the "condensed" series:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots,$$

converges. Otherwise, if the condensed series diverges, the original series also diverges.

For them condensed series of $a_n = \frac{1}{\ln n}$:

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = \sum_{k=0}^{\infty} 2^k \frac{1}{\ln 2^k} = \frac{1}{\ln 2}\sum_{k=0}^{\infty}\frac{2^k}{k},$$

We use the ratio test for sequences now if is convergent:

$$\lim_{k\to\infty}\frac{c_{k+1}}{c_k} = \lim_{k\to\infty}\frac{\frac{2^{k+1}}{k+1}}{\frac{2^k}{k}} = \lim_{k\to\infty}\frac{2^{k+1}k}{2^k(k+1)} = 2\lim_{k\to\infty}\frac{k}{k+1} = 2\lim_{k\to\infty}\frac{1}{1+\frac{1}{k}} = 2,$$

We now the condensed series diverges, for the original series also diverges.

$$(b)\sum_{n=1}^{\infty}\frac{n!}{10^n},$$

Solution :

We apply the D´Alembert Ratio Test:

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|,$$

For $a_n = \frac{n!}{10^n}$ have:

$$L = \lim_{n\to\infty}\left|\frac{\frac{(n+1)!}{10^{n+1}}}{\frac{n!}{10^n}}\right| = \lim_{n\to\infty}\left|\frac{10^n(n+1)!}{10^{n+1}n!}\right| = \lim_{n\to\infty}\left|\frac{n+1}{10}\right|,$$

Conclusion:

Since $L \to \infty$, the series diverges (and quite rapidly). In fact, the general term does not even approach zero, as shown by the limit of the ratio.

$$(c)\sum_{n=1}^{\infty}\frac{1}{2n(2n+1)},$$

Solution:

$$\sum_{n=1}^{\infty}\frac{1+0}{2n(2n+1)}=\sum_{n=1}^{\infty}\frac{2n-2n+1}{2n(2n+1)}=\sum_{n=1}^{\infty}\frac{1}{2n}-\frac{1}{2n+1},$$

Now we use the telescopic form:

$$\sum_{n=1}^{\infty}\frac{1}{2n}-\frac{1}{2n+1}=\frac{1}{2(1)}-\frac{1}{2(1)+1}+\frac{1}{2(2)}-\frac{1}{2(2)+1}+\frac{1}{2(3)}-\frac{1}{2(3)+1}+\cdots,$$

$$\sum_{n=1}^{\infty}\frac{1}{2n}-\frac{1}{2n+1}=\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}-\frac{1}{7}+\cdots,$$

Now we remember the Taylor series of $\ln(1+x)=\frac{1}{x}-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}-\frac{x^6}{6}+\frac{x^7}{7}-\cdots,$

$$\ln(2)=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\frac{1}{7}-\cdots,$$

$$1-\ln(2)=\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}-\frac{1}{7}+\cdots,$$

$$\therefore\sum_{n=1}^{\infty}\frac{1}{2n(2n+1)}=1-\ln(2)$$

The series is convergent.

$$(d)\sum_{n=1}^{\infty}\frac{1}{\sqrt{n(n+1)}},$$

Solution:

We use de the Limit Comparison Test by comparing series with harmonic series famous for be diverges $v_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\frac{1}{\sqrt{n(n+1)}}}{\frac{1}{n}} = \lim_{n\to\infty}\frac{n}{\sqrt{n^2\left(1+\frac{1}{n}\right)}} = \lim_{n\to\infty}\frac{1}{\sqrt{\left(1+\frac{1}{n}\right)}},$$

$$L = \frac{1}{\lim_{n\to\infty}\sqrt{\left(1+\frac{1}{n}\right)}} = \frac{1}{1} = 1,$$

Conclusion:

Since $0 < L < \infty$, the series shares the same behavior as the harmonic series. Therefore, it diverges.

$$(e)\sum_{n=0}^{\infty}\frac{1}{2n+1},$$

Solution:

We use the Limit Comparison Test by comparing series with harmonic series famous for be diverges $v_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\frac{1}{2n+1}}{\frac{1}{n}} = \lim_{n\to\infty}\frac{n}{2n+1} = \lim_{n\to\infty}\frac{n}{n\left(2+\frac{1}{n}\right)},$$

$$L = \lim_{n\to\infty}\frac{1}{2+\frac{1}{n}} = \frac{1}{\lim_{n\to\infty}\left(2+\frac{1}{n}\right)} = \frac{1}{2},$$

Conclusion:

Since $0 < L < \infty$, the series shares the same behavior as the harmonic series. Therefore, it diverges.

1.1.6 Test for convergence:

$$(a)\sum_{n=1}^{\infty}\frac{1}{n(n+1)},$$

Solution:

$$\sum_{n=1}^{\infty}\frac{1+0}{n(n+1)}=\sum_{n=1}^{\infty}\frac{n-n+1}{n(n+1)}=\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1},$$

Now we use the telescopic form:

$$\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1}=\frac{1}{(1)}-\frac{1}{(1)+1}+\frac{1}{(2)}-\frac{1}{(2)+1}+\frac{1}{(3)}-\frac{1}{(3)+1}+\cdots,$$

$$\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1}=1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots,$$

$$\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1}=1+\left(-\frac{1}{2}+\frac{1}{2}\right)+\left(-\frac{1}{3}+\frac{1}{3}\right)+\left(-\frac{1}{4}+\frac{1}{4}\right)+\cdots,$$

$$\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1}=1+0+0+0+\cdots,$$

$$\therefore\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+1}=1.$$

The series is convergent.

$$(b)\sum_{n=2}^{\infty}\frac{1}{n\ln(n)},$$

Solution :

Consider the function $f(x)=\frac{1}{x\ln x}$.

1. Conditions: For $2<x$, $f(x)$ is continuous, positive ,and monotonal decreasing.
2. Integral Test: The series converges if and only if the following improper integral converges:

$$I=\int_{2}^{\infty}\frac{dx}{x\ln x},$$

Evaluation :

Let $u = \ln x$, then $du = \frac{dx}{x}$.

When $x = 2, u = \ln 2$. When $x \to \infty, u \to \infty$.

$$I = \int_2^{\infty} \frac{du}{u} = [\ln u]_0^{\infty},$$

$$I = \lim_{b\to\infty}(\ln(2) + \ln(b)) = \ln(2) + \lim_{b\to\infty}(\ln(b)),$$

Conclusion:

Since the integral diverges , the series diverges.

$$(c) \sum_{n=1}^{\infty} \frac{1}{n2^n},$$

Solution :

We apply the D´Alembert Ratio Test:

$$L = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|,$$

For $a_n = \frac{1}{n2^n}$ have:

$$L = \lim_{n\to\infty} \left|\frac{\frac{1}{(n+1)2^{n+1}}}{\frac{1}{n2^n}}\right| = \lim_{n\to\infty} \left|\frac{n2^n}{(n+1)2^{n+1}}\right| = \lim_{n\to\infty} \left|\frac{n}{2(n+1)}\right| = \lim_{n\to\infty} \left|\frac{1}{2}\right| = \frac{1}{2},$$

Conclusion:

Since $0 < L < 1$, the series converges.

$$(d) \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right),$$

Solution :

We use the Limit Comparison Test by comparing series with harmonic series famous for be diverges $v_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\ln\left(1 + \frac{1}{n}\right)}{\frac{1}{n}} = \lim_{n\to\infty} n \ln\left(1 + \frac{1}{n}\right) = \lim_{n\to\infty} \left[\ln\left(1 + \frac{1}{n}\right)^n\right],$$

$$L = \ln\left[\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n\right] = \ln[e] = 1,$$

Conclusion:

Since $0 < L < \infty$, the series shares the same behavior as the harmonic series. Therefore, it diverges.

$$(e)\sum_{n=1}^{\infty}\frac{1}{n\cdot n^{\frac{1}{n}}},$$

Solution :

We use the Limit Comparison Test by comparing series with harmonic series famous for be diverges $v_n = \frac{1}{n}$:

$$L = \lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\frac{1}{n\cdot n^{\frac{1}{n}}}}{\frac{1}{n}} = \lim_{n\to\infty}\frac{n}{n\cdot n^{\frac{1}{n}}} = \lim_{n\to\infty}\frac{1}{n^{\frac{1}{n}}},$$

$$L = \frac{1}{\lim\limits_{n\to\infty} n^{\frac{1}{n}}} = \frac{1}{1} = 1,$$

Conclusion:

Since $0 < L < \infty$, the series shares the same behavior as the harmonic series. Therefore, it diverges.

1.1.7 For what values of p and q will $\sum_{n=2}^{\infty}\frac{1}{n^p(\ln n)^q}$ converge ?

Solution :

Consider the function $f(x)=\frac{1}{x^p(\ln x)^q}$.

1. Conditions: For $1<x$, $f(x)$ is continuous, positive ,and monotonal decreasing.
2. Integral Test: The series converges if and only if the following improper integral converges:

$$I=\int_2^{\infty}\frac{dx}{x^p(\ln x)^q},$$

Evaluation :

Let $u=\ln x$, then $du=\frac{dx}{x}$ or $dx=e^u du$.

When $x=2, u=\ln 2$. When $x\to\infty, u\to\infty$.

$$I=\int_{\ln 2}^{\infty}\frac{e^{-up}e^u du}{u^q}=\int_{\ln 2}^{\infty}e^{u(1-p)}u^{-q}du,$$

If $p=1$ have :

$$I=\int_{\ln 2}^{\infty}u^{-q}du=\left[\frac{u^{-q+1}}{1-q}\right]_{\ln 2}^{\infty}=\lim_{b\to\infty}\left[\frac{b^{-q+1}}{1-q}-\frac{\ln(2)^{-q+1}}{1-q}\right],$$

For limit we now know if $q>1$ the integral and series converges, if $q\leq 1$ diverges.

Remember $E_q(z)$:

$$E_q(z)=\int_z^{\infty}e^{-zt}t^{-q}dt,$$

$$I=\int_{\ln 2}^{\infty}e^{u(1-p)}u^{-q}du=(\ln 2)^{1-q}E_q\big((p-1)\ln(2)\big),$$

For $p>1$ the integral converges for any q, For $p<1$ the integral diverges for any q,

Conclusion:

The series convergent for $\begin{cases}p>1, & all\ q,\\ p=1, & q>1\end{cases}$ divergent for $\begin{cases}p<1, & all\ q,\\ p=1, & q\leq 1\end{cases}$.

1.1.8 Given $\sum_{n=1}^{1,000} n^{-1} = 7.485\,470 \ldots$ set upper and lower bounds on the Euler-Mascheroni constant.

Solution :

Remember the Euler-Mascheroni constant and bounds:

$$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \ln(n)\right),$$

$$\sum_{n=1}^{N}\frac{1}{n} - \ln(N+1) < \gamma < \sum_{n=1}^{N}\frac{1}{n} - \ln(N),$$

We have $N = 1,000$ for them :

$$\sum_{n=1}^{1,000}\frac{1}{n} - \ln(1,000+1) = 7.485\,470 - 6.908\,754 = 0.5767,$$

$$\sum_{n=1}^{1,000}\frac{1}{n} - \ln(1,000) = 7.485\,470 - 6.907\,755 = 0.5778,$$

$$\therefore 0.5767 < \gamma < 0.5778$$

1.1.9 (From **Olbers' paradox**.) Assume a static universe in which the stars are uniformly distributed. Divide all space into shells of constant thickness; the stars in any one shell by themselves subtend a solid angle of ω0. **Allowing for the blocking out of distant stars by nearer stars**, show that the total net solid angle subtended by all stars, shells extending to infinity, is **exactly** 4π. [Therefore the night sky should be ablaze with light. For more details, see E. Harrison, *Darkness at Night: A Riddle of the Universe.* Cambridge, MA: Harvard University Press (1987).]

Solution :

The Physical Model:

Assume a uniform distribution of stars with number density n and cross-sectional area A. We divide space into concentric shells of thickness dr. The probability that a line of sight is intercepted by a star in a shell at distance r is proportional to the area covered by the stars in that shell.

The Differential Blocking Effect:

Let $P(r)$ be the probability that a line of sight has hit a star up to distance r. When we cross an additional thickness dr, the change in probability dP is proportional to the current "clear" probability $P(r)$ times the area density of the stars in that new layer ($nAdr$):

$$-dP = (P(r) - 1) \cdot nAdr,$$

Solving the Differential Equation:

This is a standard first-order decay equation:

$$\frac{dP}{P-1} = -nAdr \Longrightarrow P(r) = 1 - e^{-nAr}C,$$

$$P(0) = 1 - e^{-nA(0)}C \Longrightarrow C = 1,$$

$$\Longrightarrow P(r) = 1 - e^{-nAr},$$

Total Subtended Solid Angle:

The fraction of the sky covered by stars, f, is the complement of the probability of finding a clear path to infinity, since the total solid angle of the celestial sphere is 4π, the net solid angle subtended by all stars is:

$$\Omega_{net} = 4\pi \cdot \lim_{r\to\infty} f(r) = 4\pi \cdot \lim_{r\to\infty}[1 - e^{-nAr}] = 4\pi.$$

1.1.10 Test for convergence:

$$\sum_{n=1}^{\infty}\left[\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}\right]^2=\frac{1}{4}+\frac{9}{64}+\frac{25}{256}+\cdots.$$

Solution:

Remember de Gauss' test:

$$\frac{u_n}{u_{n+1}}=1+\frac{h}{n}+\frac{B(n)}{n^2},$$

Where $B(n)$ is bounded function ,for then series is:

Converge if $h>1$, diverge if $h\leq 1$.

Now :

$$\frac{u_n}{u_{n+1}}=\frac{\left[\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}\right]^2}{\left[\frac{1\cdot 3\cdot 5\cdots(2(n+1)-1)}{2\cdot 4\cdot 6\cdots(2(n+1))}\right]^2}=\frac{\left[\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}\right]^2}{\left[\frac{1\cdot 3\cdot 5\cdots(2n+1)}{2\cdot 4\cdot 6\cdots(2n+2)}\right]^2}=\left[\frac{2n+2}{2n+1}\right]^2,$$

$$\frac{u_n}{u_{n+1}}=\left[1+\frac{1}{n}\right]^2\left[1+\frac{1}{2n}\right]^{-2}$$

We use the Taylor expansion $\left(1+\frac{1}{x}\right)^2=1+\frac{2}{x}+\frac{1}{x^2}+\cdots,\left(1+\frac{1}{2x}\right)^{-2}=1-\frac{1}{x}+\frac{3}{4x^2}+\cdots;$

$$\frac{u_n}{u_{n+1}}=\left[1+\frac{2}{n}+\frac{1}{n^2}+\cdots\right]\left[1-\frac{1}{n}+\frac{3}{4n^2}+\cdots,\right],$$

$$\frac{u_n}{u_{n+1}}=1+\left(\frac{2}{n}-\frac{1}{n}\right)+\left(\frac{3}{4n^2}+\frac{1}{n^2}-\frac{2}{n^2}\right)+\cdots,$$

$$\frac{u_n}{u_{n+1}}=1+\frac{1}{n}+\frac{-\frac{1}{4}}{n^2}+\cdots,$$

Is easy now identify $h=1$ and $B(n)=-\frac{1}{4}+O(n^{-3})$ which is a bounded function; therefore, the series diverges.

1.1.11 Determine whether each of these series is convergent, and if so, whether it is absolutely convergent:

$$(a)\ \frac{\ln(2)}{2}-\frac{\ln(3)}{3}+\frac{\ln(4)}{4}-\frac{\ln(5)}{5}+\frac{\ln(6)}{6}-\cdots,$$

Solution:

First we use The Leibniz criterion , applies to series of the form:

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n,$$

Whit $a_n > 0$, and states that if a_n is monotonically decreasing (for sufficiently large n) and $\lim_{n\to\infty} a_n = 0$, then the series converges.

$$a_n = \frac{\ln n}{n},$$

If $a_{n+1} < a_n$ is monotonically decreasing (for sufficiently large n):

$$\frac{\ln(n+1)}{n+1} < \frac{\ln n}{n},$$

$$n\ln(n+1) < (n+1)\ln n,$$

$$\ln(n+1)^n < \ln n^{n+1},$$

$$(n+1)^n < n^{n+1},$$

$$\left(1+\frac{1}{n}\right)^n < n^n,$$

When sufficiently large n :

$$e < n \Longrightarrow \therefore a_{n+1} < a_n | n \geq 3,$$

For other side :

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\frac{\ln n}{n} = \frac{\lim_{n\to\infty}\left\{\frac{d}{dn}[\ln n]\right\}}{\lim_{n\to\infty}\left\{\frac{d}{dn}n\right\}} = \frac{\lim_{n\to\infty}\left\{\frac{1}{n}\right\}}{\lim_{n\to\infty}\{1\}} = 0,$$

Then the series converges for $n \geq 3$.

We use the Limit Comparison Test by comparing the series of absolute values with harmonic series famous for be diverges $v_n = \frac{1}{n}, u_n = \frac{\ln n}{n}$:

$$L = \lim_{n\to\infty}\left|\frac{u_n}{v_n}\right| = \lim_{n\to\infty}\frac{\frac{\ln n}{n}}{\frac{1}{n}} = \lim_{n\to\infty}\ln n,$$

Since $L \to \infty$, the series shares the same behavior as the harmonic series. Therefore, it diverges.

Conclusion :

The series is Conditional convergent.

$$(b)\ \frac{1}{1}+\frac{1}{2}-\frac{1}{3}-\frac{1}{4}+\frac{1}{5}+\frac{1}{6}-\frac{1}{7}-\frac{1}{8}+\cdots,$$

Solution :

First we can group:

$$\left(\frac{1}{1}+\frac{1}{2}\right)-\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}\right)-\left(\frac{1}{7}+\frac{1}{8}\right)+\cdots,$$

And now is easy to see:

$$a_n = \frac{1}{2n-1}+\frac{1}{2n} = \frac{4n-1}{(2n-1)2n},$$

We use The Leibniz criterion , applies to series of the form:

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n,$$

Whit $a_n > 0$, and states that if a_n is monotonically decreasing (for sufficiently large n) and $\lim_{n\to\infty} a_n = 0$, then the series converges.

If $a_{n+1} < a_n$ is monotonically decreasing (for sufficiently large n):

$$2(n+1)-1 > 2n-1 \wedge 2(n+1) > 2n,$$

$$\frac{1}{2(n+1)-1}+\frac{1}{2(n+1)} < \frac{1}{2n-1}+\frac{1}{2n},$$

$$\therefore a_{n+1} < a_n.$$

For other side :

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left[\frac{1}{2n-1}+\frac{1}{2n}\right] = \lim_{n\to\infty}\left[\frac{1}{2n-1}\right]+\lim_{n\to\infty}\left[\frac{1}{2n}\right] = 0+0 = 0,$$

For absolute convergence:

$$\sum_{n=1}^{\infty}|a_n| = \frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\cdots,$$

This is the harmonic series and divergent therefore, the series is Conditional convergent.

$$(c)\,\frac{1}{1}-\frac{1}{2}-\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}-\frac{1}{7}-\frac{1}{8}-\frac{1}{9}-\frac{1}{10}+\frac{1}{11}\ldots+\frac{1}{15}-\frac{1}{16}\ldots-\frac{1}{21}+\cdots.$$

Solution :

Pattern Observation:

The series groups terms with the same sign in blocks of increasing size (1,2,3, ...).

$$S=\left(\frac{1}{1}\right)-\left(\frac{1}{2}+\frac{1}{3}\right)+\left(\frac{1}{4}+\frac{1}{5}+\frac{1}{6}\right)-\left(\frac{1}{7}+\frac{1}{8}+\frac{1}{9}+\frac{1}{10}\right)+\cdots,$$

Theoretical Analysis:

Let B_k be the sum of the k-th block of terms. The number of terms in the k-th block is k. Using the integral approximation for the harmonic sum, the magnitude of each block can be estimated as:

$$|B_k| = \sum_{n=N_k}^{N_k+k-1}\frac{1}{n} \approx \ln\left(\frac{N_k+k}{N_k}\right),$$

Where N_k is the starting index of the k-th block.

As the blocks grow in size, their contribution does not vanish. In fact, for a series to satisfy the Cauchy Criterion for convergence, the sum of any tail of terms must be arbitrarily small. However, in this series, the partial sums S continue to fluctuate with an amplitude that does not tend to zero because the blocks of same-signed terms are large enough to counteract the decay of $\frac{1}{n}$.

Conclusion:

Since the blocks of same-signed terms grow in size, the oscillations of the partial sums do not diminish. This violates the necessary conditions for convergence.

The series diverges.

1.1.12 Catalan's constant $\beta(2)$ is defined by:

$$\beta(2) = \sum_{k=0}^{\infty}(-1)^k(2k+1)^{-2} = \frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} + \cdots.$$

Calculate $\beta(2)$ to six-digit accuracy.

Solution:

Acceleration via Kummer's Transformation:

To avoid the "brute force" summation, we can link $\beta(2)$ to faster-converging series. One of the most elegant representations involves the following identity (derived from the generalized hypergeometric function or the Clausen function):

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\sum_{n=0}^{\infty}\frac{1}{\binom{2n}{n}(2n+1)^2},$$

For $n = 0$:

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8} \approx 0.517482 + 0.375 \approx 0.892482$$

For $n = 1$:

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\left(1+\frac{1}{9}\right) \approx 0.517482 + 0.375(1+0.055555) \approx 0.913315,$$

For $n = 2$:

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\left(1+\frac{1}{18}+\frac{1}{300}\right) \approx 0.915501,$$

For $n = 3$:

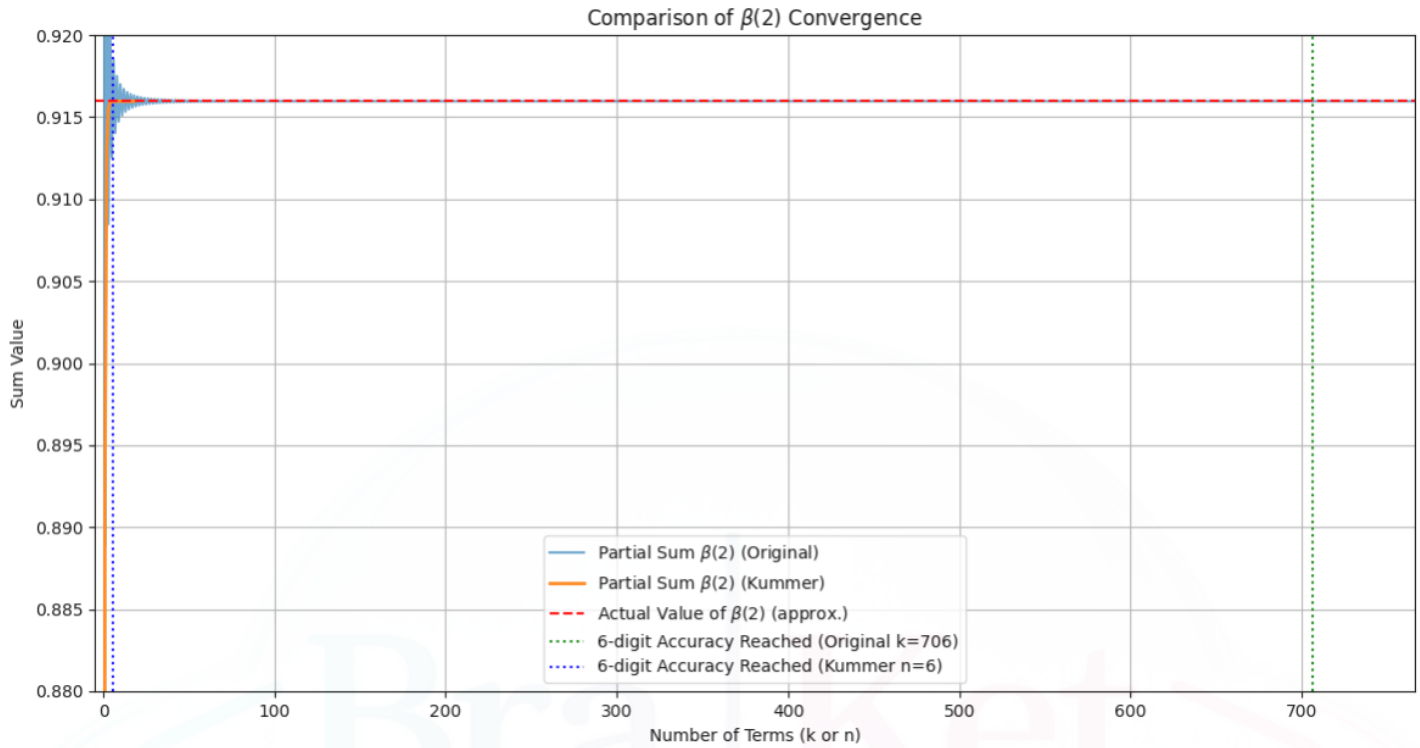
$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\left(1+\frac{1}{18}+\frac{1}{300}+\frac{1}{20\cdot 7^2}\right) \approx 0.915584,$$

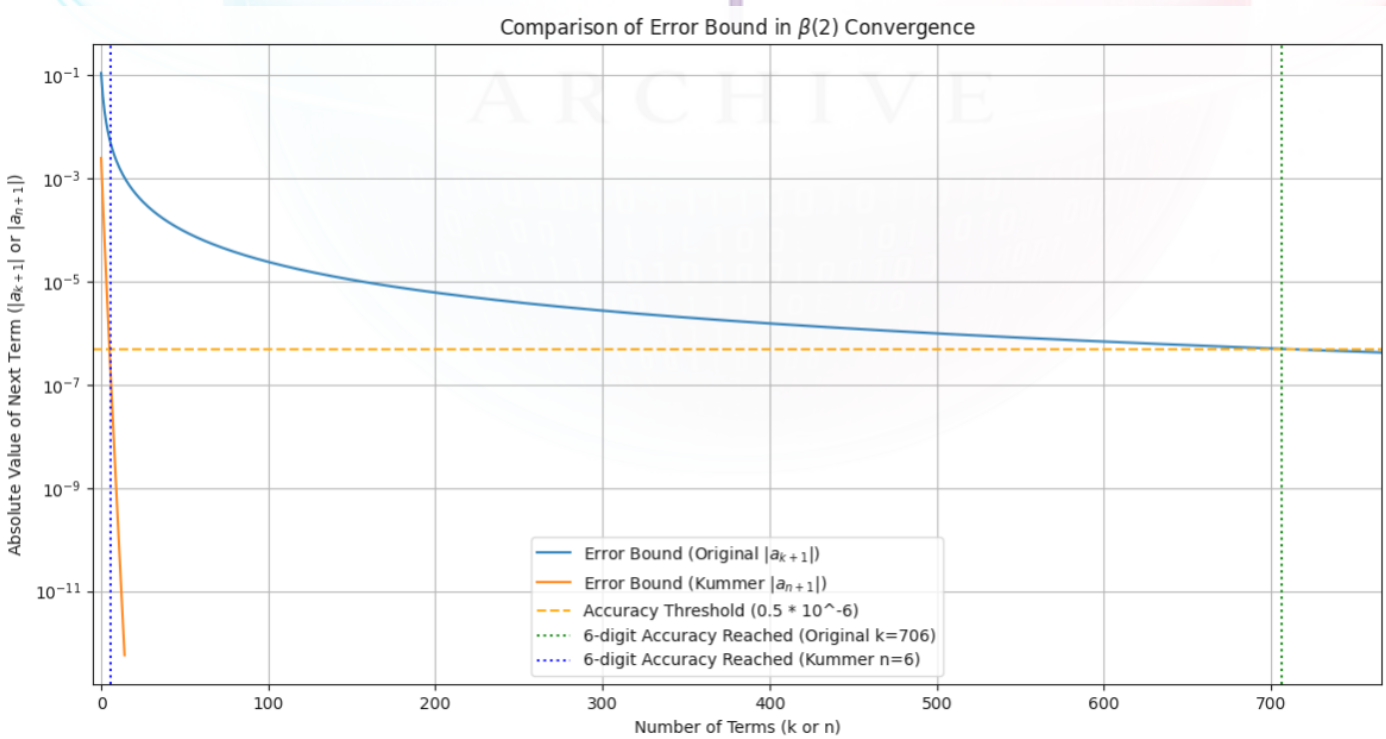
For $n = 4$:

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\left(1+\frac{1}{18}+\frac{1}{300}+\frac{1}{20\cdot 7^2}+\frac{1}{70\cdot 9^2}\right) \approx 0.915595,$$

For $n = 5$:

$$\beta(2) = \frac{\pi}{8}\ln\left(2+\sqrt{3}\right) + \frac{3}{8}\left(1+\frac{1}{18}+\frac{1}{300}+\frac{1}{20\cdot 7^2}+\frac{1}{70\cdot 9^2}+\frac{1}{252\cdot 11^2}\right) \approx 0.915596,$$


Comparison of $\beta(2)$ Convergence
Sum Value
Number of Terms (k or n)
Partial Sum $\beta(2)$ (Original)
Partial Sum $\beta(2)$ (Kummer)
Actual Value of $\beta(2)$ (approx.)
6-digit Accuracy Reached (Original k=706)
6-digit Accuracy Reached (Kummer n=6)


Comparison of Error Bound in $\beta(2)$ Convergence
Absolute Value of Next Term ($|a_{k+1}|$ or $|a_{n+1}|$)
Number of Terms (k or n)
Error Bound (Original $|a_{k+1}|$)
Error Bound (Kummer $|a_{n+1}|$)
Accuracy Threshold (0.5 * 10^-6)
6-digit Accuracy Reached (Original k=706)
6-digit Accuracy Reached (Kummer n=6)

Colab sim:

https://colab.research.google.com/drive/1GQHqG6wLf91bvyOwKlIbL4lbGqy8iywr?usp=sharing

$\beta(2)$ is equivalent to the following definite integral:

$$\beta(2) = \int_0^1 \frac{\arctan(x)}{x} dx = 0.915966,$$

We recognize this integral as the standard definition of the Catalan Constant (G).

1.1.13 Show how to combine $\zeta(2) = \sum_{n=1}^{\infty} n^{-2}$ with $\alpha_1$and α_2 to obtain a series converging as n^{-4}.

Solution :

Remember :

$$\alpha_1 = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1,$$

$$\alpha_2 = \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \frac{1}{4},$$

$$\zeta(2) = \sum_{n=1}^{\infty} n^{-2} = \frac{\pi^2}{6}$$

Now we can combine :

$$\zeta(2) - \alpha_1 = \sum_{n=1}^{\infty} n^{-2} - \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n^2} - \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)},$$

$$\zeta(2) - \alpha_1 - \alpha_2 = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)} - \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)},$$

$$\zeta(2) - \alpha_1 - \alpha_2 = \sum_{n=1}^{\infty} \frac{n(n+1)(n+2) - n^2(n+1)}{n^2(n+1)n(n+1)(n+2)},$$

$$\zeta(2) - \alpha_1 - \alpha_2 = \sum_{n=1}^{\infty} \frac{(n+1)(n(n+2) - n^2)}{n^2(n+1)n(n+1)(n+2)},$$

$$\zeta(2) - \alpha_1 - \alpha_2 = \sum_{n=1}^{\infty} \frac{2}{n^2(n+1)(n+2)},$$

Now is trivial to now the converging as n^{-4} is:

$$\therefore \sum_{n=1}^{\infty} \frac{2}{n^2(n+1)(n+2)} = \zeta(2) - \alpha_1 - \alpha_2 = \frac{\pi^2}{6} - 1 - \frac{1}{4} \approx 0.3949.$$

1.1.14 Given a method of computing :

$$\lambda(3) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3},$$

that converges at least as fast as n^{-8} and obtain a result good to six decimal places.

Solution :

Remember :

$$\alpha = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)(2n+3)} = \sum_{n=1}^{\infty} \frac{4}{4}\left[\frac{1}{(2n-1)(2n+1)(2n+3)}\right],$$

$$\alpha = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)(2n+3)} = \sum_{n=1}^{\infty} \frac{1}{4}\left[\frac{4+0}{(2n-1)(2n+1)(2n+3)}\right],$$

$$\alpha = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)(2n+3)} = \sum_{n=1}^{\infty} \frac{1}{4}\left[\frac{(2n+3)-(2n-1)}{(2n-1)(2n+1)(2n+3)}\right],$$

$$\alpha = \sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)(2n+3)} = \frac{1}{4}\sum_{n=1}^{\infty}\left[\frac{1}{(2n-1)(2n+1)} - \frac{1}{(2n+1)(2n+3)}\right],$$

Now we use the telescopic form:

$$\alpha = \frac{1}{4}\left\{\left[\frac{1}{(2(1)-1)(2(1)+1)} - \frac{1}{(2(1)+1)(2(1)+3)}\right] + \left[\frac{1}{(2(2)-1)(2(2)+1)} - \frac{1}{(2(2)+1)(2(2)+3)}\right] + \cdots\right\},$$

$$\alpha = \frac{1}{4}\left\{\left[\frac{1}{(1)(3)} - \frac{1}{(3)(5)}\right] + \left[\frac{1}{(3)(5)} - \frac{1}{(5)(7)}\right] + \cdots\right\},$$

$$\alpha = \frac{1}{4}\left\{\frac{1}{3} + \left[\frac{1}{15} - \frac{1}{15}\right] + \left[\frac{1}{35} - \frac{1}{35}\right] + \cdots\right\},$$

$$\alpha = \frac{1}{4}\left\{\frac{1}{3} + [0] + [0] + \cdots\right\},$$

$$\therefore \alpha = \frac{1}{12}.$$

Now we can combine :

$$\lambda(3)-\alpha=\sum_{n=0}^{\infty}\frac{1}{(2n+1)^3}-\sum_{n=1}^{\infty}\frac{1}{(2n-1)(2n+1)(2n+3)},$$

$$\lambda(3)-\alpha=1+\sum_{n=1}^{\infty}\frac{1}{(2n+1)^3}-\frac{1}{(2n-1)(2n+1)(2n+3)},$$

$$\lambda(3)-\alpha=1+\sum_{n=1}^{\infty}\frac{(2n-1)(2n+1)(2n+3)-(2n+1)^3}{(2n+1)^3(2n-1)(2n+1)(2n+3)},$$

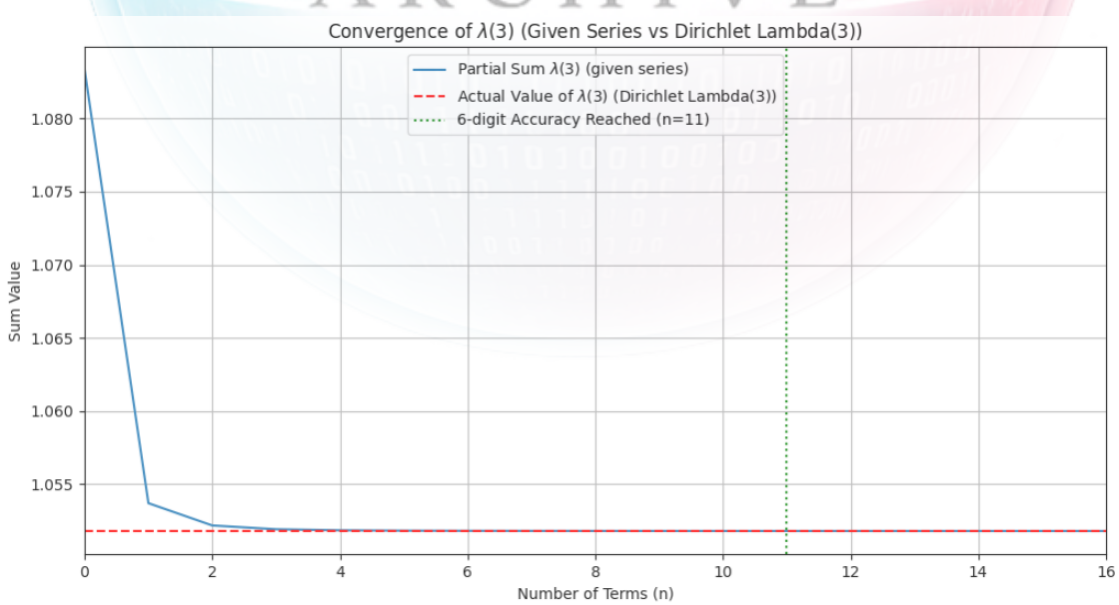
$$\lambda(3)-\alpha=1+\sum_{n=1}^{\infty}\frac{(2n-1)(2n+3)-(2n+1)^2}{(2n+1)^3(2n-1)(2n+3)},$$

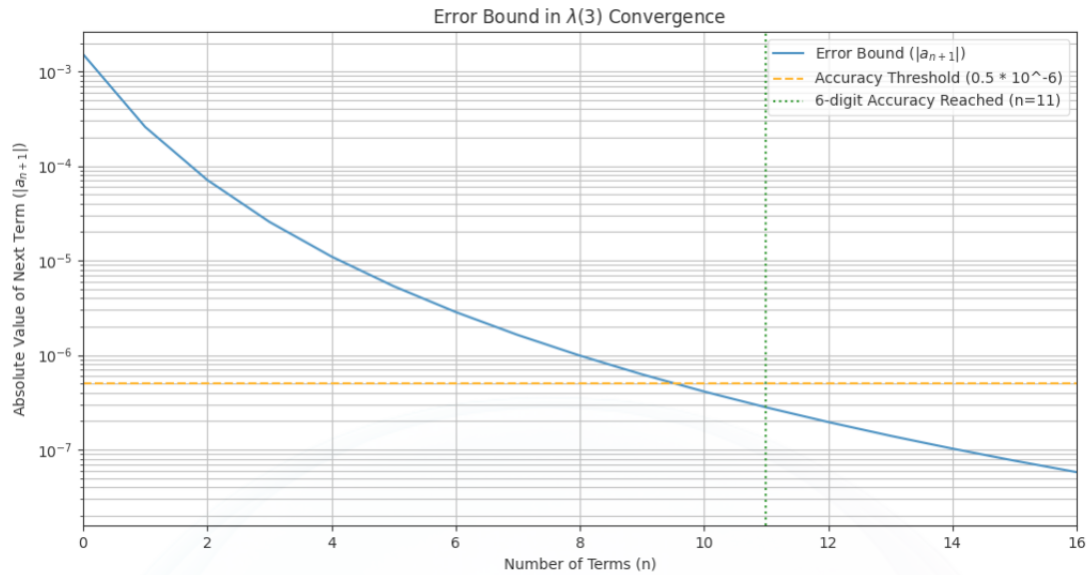
$$\lambda(3)-\alpha=1+\sum_{n=1}^{\infty}\frac{4n^2+6n-2n-3-(4n^2+4n+1)}{(2n+1)^3(2n-1)(2n+3)},$$

$$\lambda(3)-\alpha=1+\sum_{n=1}^{\infty}\frac{-4}{(2n+1)^3(2n-1)(2n+3)},$$

We can computing this , that converges at n^{-5} :

$$\lambda(3)=1+\frac{1}{12}-4\sum_{n=1}^{\infty}\frac{1}{(2n+1)^3(2n-1)(2n+3)}\approx 1.051800,$$





1.1.15 Show that:

$$(a)\sum_{n=2}^{\infty}[\zeta(n)-1]=1,(b)\sum_{n=2}^{\infty}(-1)^n[\zeta(n)-1]=\frac{1}{2},$$

Where $\zeta(n)$ is the Riemann zeta function.

Solution :

Recall $\zeta(n)=\sum_{k=1}^{\infty}\frac{1}{k^n}=1+\frac{1}{2^n}+\frac{1}{3^n}+\cdots;$

$$\zeta(n)-1=\sum_{k=2}^{\infty}\frac{1}{k^n},$$

Substitute this into the original series:

$$S=\sum_{n=2}^{\infty}\sum_{k=2}^{\infty}\frac{1}{k^n},$$

Since all terms are positive, we can swap the summation order (Fubini's Theorem for series) :

$$S=\sum_{k=2}^{\infty}\sum_{n=2}^{\infty}\left(\frac{1}{k}\right)^n,$$

The inner sum is a geometric series starting at $n=2$ with ratio $r=\frac{1}{k}$:

$$\sum_{n=2}^{\infty}\left(\frac{1}{k}\right)^n=\frac{\left(\frac{1}{k}\right)^2}{1-\frac{1}{k}}=\left(\frac{1}{k}\right)^2\left(\frac{k}{k-1}\right)=\frac{1}{k(k-1)},$$

Now we sum over k:

$$S=\sum_{k=2}^{\infty}\frac{1}{k(k-1)}=\sum_{k=2}^{\infty}\frac{1-k+k}{k(k-1)}=\sum_{k=2}^{\infty}\frac{1}{k-1}-\frac{1}{k},$$

Now we use the telescopic form:

$$\sum_{k=2}^{\infty}\frac{1}{k-1}-\frac{1}{k}=\frac{1}{(2)-1}-\frac{1}{(2)}+\frac{1}{(3)-1}-\frac{1}{(3)}+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k-1}-\frac{1}{k}=1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k-1}-\frac{1}{k}=1+\left(-\frac{1}{2}+\frac{1}{2}\right)+\left(-\frac{1}{3}+\frac{1}{3}\right)+\left(-\frac{1}{4}+\frac{1}{4}\right)+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k-1}-\frac{1}{k}=1+0+0+0+\cdots,$$

$$\therefore\sum_{n=2}^{\infty}[\zeta(n)-1]=1.$$

For (b):

$$S=\sum_{n=2}^{\infty}\sum_{k=2}^{\infty}\frac{(-1)^n}{k^n}=\sum_{k=2}^{\infty}\sum_{n=2}^{\infty}\left(\frac{-1}{k}\right)^n,$$

The inner sum is a geometric series starting at $n=2$ with ratio $r=-\frac{1}{k}$:

$$\sum_{n=2}^{\infty}\left(\frac{1}{k}\right)^n=\frac{\left(-\frac{1}{k}\right)^2}{1+\frac{1}{k}}=\left(\frac{1}{k}\right)^2\left(\frac{k}{k+1}\right)=\frac{1}{k(k+1)},$$

Now we sum over k:

$$S=\sum_{k=2}^{\infty}\frac{1}{k(k+1)}=\sum_{k=2}^{\infty}\frac{1-k+k}{k(k+1)}=\sum_{k=2}^{\infty}\frac{1}{k}-\frac{1}{k+1},$$

Now we use the telescopic form:

$$\sum_{k=2}^{\infty}\frac{1}{k}-\frac{1}{k+1}=\frac{1}{(2)}-\frac{1}{(2)+1}+\frac{1}{(3)}-\frac{1}{(3)+1}+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k}-\frac{1}{k+1}=\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k}-\frac{1}{k+1}=\frac{1}{2}+\left(-\frac{1}{3}+\frac{1}{3}\right)+\left(-\frac{1}{4}+\frac{1}{4}\right)+\cdots,$$

$$\sum_{k=2}^{\infty}\frac{1}{k}-\frac{1}{k+1}=\frac{1}{2}+0+0+0+\cdots,$$

$$\therefore\sum_{n=2}^{\infty}(-1)^n[\zeta(n)-1]=\frac{1}{2}.$$

1.1.16 The convergence improvement of 1.1.11 may be carried out more expediently (in this special case) by putting α_2 from Eq:

$$\alpha_p=\sum_{n=1}^{\infty}\frac{1}{n(n+1)\cdots(n+p)}=\frac{1}{pp!},$$

Into a more symmetric form: Replacing n by $n-1$, we have:

$$\alpha_2^{'}=\sum_{n=2}^{\infty}\frac{1}{(n-1)n(n+1)}=\frac{1}{4},$$

(a) Combine $\zeta(3)$ and $\alpha_2^{'}$ to obtain convergence as n^{-5}.

Solution :

Recall $\zeta(n)=\sum_{k=1}^{\infty}\frac{1}{k^n}=1+\frac{1}{2^n}+\frac{1}{3^n}+\cdots;$

$$\zeta(3)=\sum_{n=1}^{\infty}\frac{1}{n^3},$$

Now we can combine :

$$\zeta(3)-\alpha_2^{'}=\sum_{n=1}^{\infty}\frac{1}{n^3}-\sum_{n=2}^{\infty}\frac{1}{(n-1)n(n+1)}=1+\sum_{n=2}^{\infty}\frac{1}{n^3}-\frac{1}{(n-1)n(n+1)},$$

$$\zeta(3)-\alpha_2^{'}=1+\sum_{n=2}^{\infty}\frac{(n-1)n(n+1)-n^3}{n^3(n-1)n(n+1)}=1+\sum_{n=2}^{\infty}\frac{n^3-n-n^3}{n^3(n-1)n(n+1)}$$

$$\zeta(3)-\alpha_2^{'}=1-\sum_{n=2}^{\infty}\frac{1}{n^3(n-1)(n+1)},$$

Now is trivial to now the converging as n^{-5} is:

$$\therefore \zeta(3) = \frac{5}{4} - \sum_{n=2}^{\infty} \frac{1}{n^3(n-1)(n+1)} = \frac{5}{4} - \sum_{n=2}^{\infty} \frac{1}{n^3(n^2-1)},$$

(b) Let $\alpha_4^{'}$ be α_4 with $n \longrightarrow n-2$.Combine $\zeta(3)$, $\alpha_2^{'}$ to obtain convergence as n^{-7}.

Solution :

Remember :

$$\alpha_4 = \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)(n+4)} = \frac{1}{4 \cdot 4!} = \frac{1}{96},$$

Let $\alpha_4^{'}$ be α_4 with $n \longrightarrow n-2$:

$$\alpha_4 = \sum_{n=3}^{\infty} \frac{1}{(n-2)\big((n-2)+1\big)\big((n-2)+2\big)\big((n-2)+3\big)\big((n-2)+4\big)},$$

$$\alpha_4^{'} = \sum_{n=3}^{\infty} \frac{1}{(n-2)(n-1)n(n+1)(n+2)} = \frac{1}{96},$$

Now we can combine :

$$\zeta(3) - \alpha_2^{'} + \alpha_4^{'} = 1 - \sum_{n=2}^{\infty} \frac{1}{n^3(n-1)(n+1)} + \sum_{n=3}^{\infty} \frac{1}{(n-2)(n-1)n(n+1)(n+2)},$$

$$\zeta(3) - \alpha_2^{'} + \alpha_4^{'} = 1 - \frac{1}{24} - \sum_{n=3}^{\infty} \frac{1}{n^3(n-1)(n+1)} - \frac{1}{(n-2)(n-1)n(n+1)(n+2)},$$

$$\zeta(3) - \alpha_2^{'} + \alpha_4^{'} = 1 - \frac{1}{24} - \sum_{n=3}^{\infty} \frac{(n-2)(n-1)n(n+1)(n+2) - n^3(n-1)(n+1)}{n^3(n-1)(n+1)(n-2)(n-1)n(n+1)(n+2)},$$

$$\zeta(3) - \alpha_2^{'} + \alpha_4^{'} = 1 - \frac{1}{24} - \sum_{n=3}^{\infty} \frac{\big((n-2)n(n+2) + n^3\big)(n-1)(n+1)}{n^3(n-1)(n+1)(n-2)(n-1)n(n+1)(n+2)},$$

$$\zeta(3) - \alpha_2^{'} + \alpha_4^{'} = 1 - \frac{1}{24} - \sum_{n=3}^{\infty} \frac{n^3 - 4n - n^3}{n^3(n-1)(n+1)(n-2)n(n+2)},$$

$$\zeta(3) - \alpha_2' + \alpha_4' = 1 - \frac{1}{24} - 4\sum_{n=3}^{\infty} \frac{1}{n^3(n-1)(n+1)(n-2)(n+2)},$$

Now is trivial to now the converging as n^{-7} is:

$$\zeta(3) = 1 + \frac{1}{4} - \frac{1}{96} - \frac{1}{24} - 4\sum_{n=3}^{\infty} \frac{1}{n^3(n^2-1)(n^2-4)},$$

$$\therefore \zeta(3) = \frac{115}{96} + 4\sum_{n=3}^{\infty} \frac{1}{n^3(n^2-1)(n^2-4)},$$

(c) If $\zeta(3)$ is to be calculated to six-decimal place accuracy (error 5×10^{-7}), how many terms are required for $\zeta(3)$ alone ? combined as in part (a)? combined as in part (b)?
Note. The error may be estimated using the corresponding integral.

Solution :

Remember the error estimated integral:

$$R_n \approx \int_n^{\infty} f(x)dx,$$

For $\zeta(3)$ with error 5×10^{-7} have :

$$R_n \approx \int_n^{\infty} f(x)dx = \int_n^{\infty} x^{-3}dx = \left[\frac{x^{-2}}{-2}\right]_n^{\infty} = \frac{1}{2n^2},$$

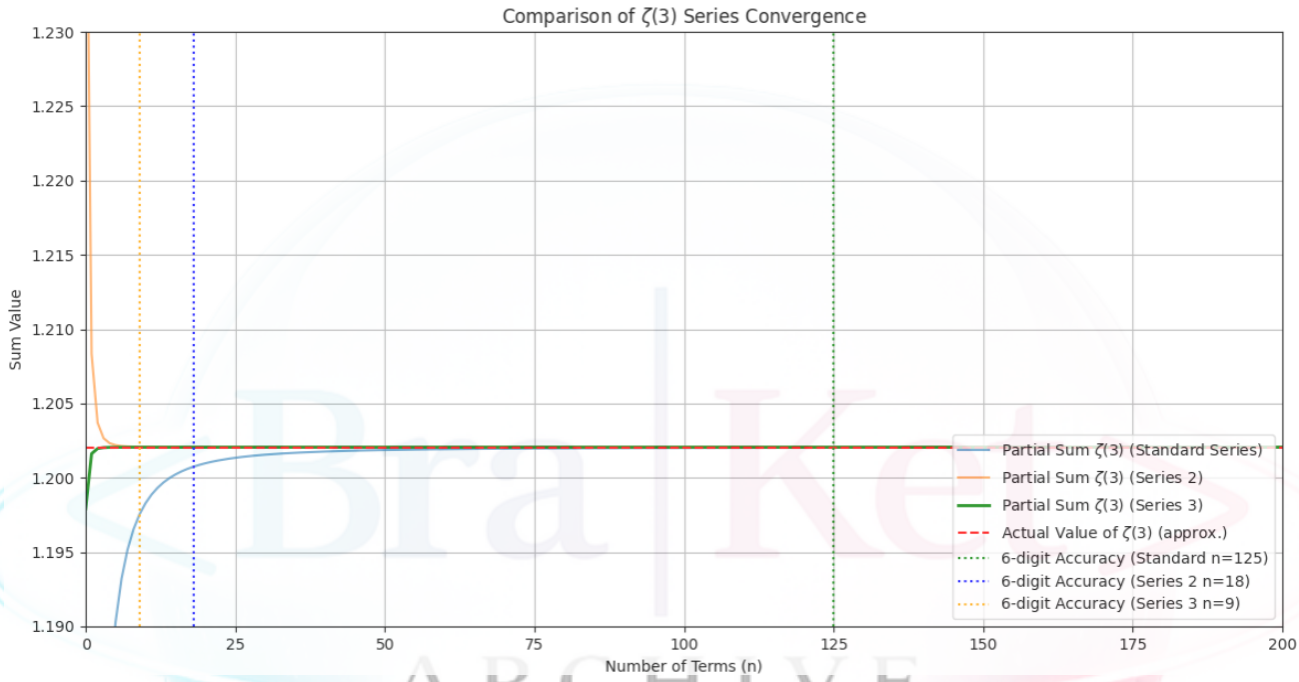
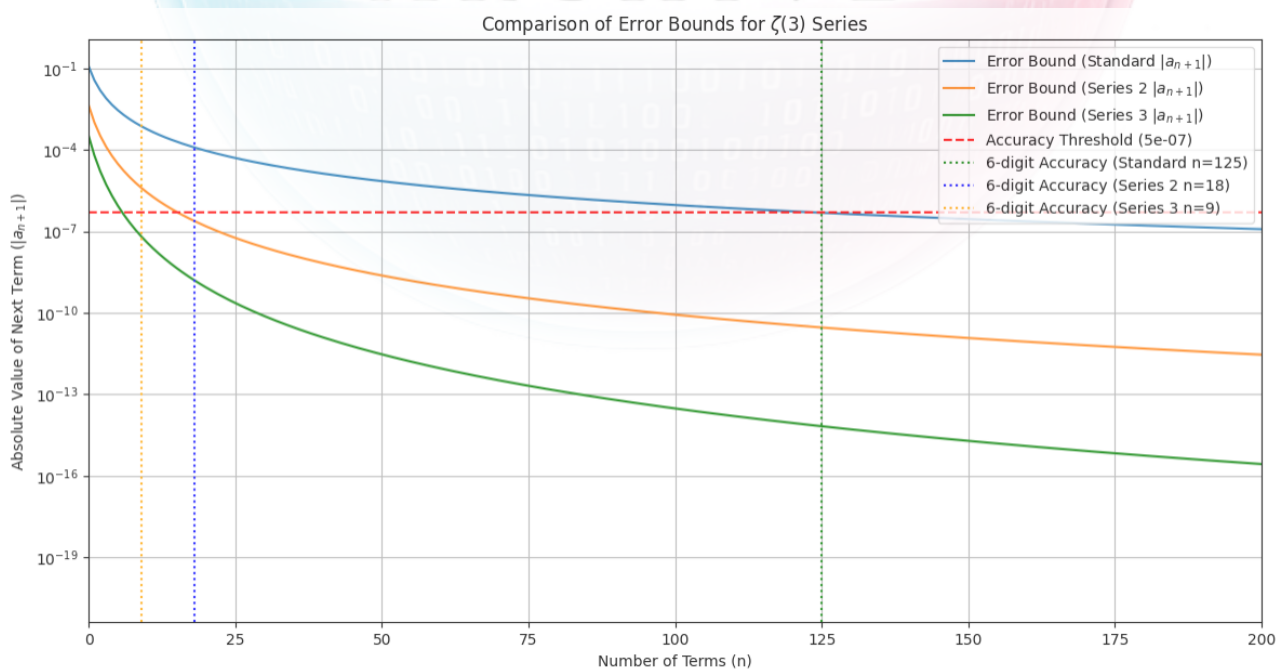
$$\frac{1}{2n^2} \leq 5 \times 10^{-7} \Longrightarrow n^2 \geq \frac{1}{10} \times 10^7 \Longrightarrow n \geq 1{,}000\,,$$

For (a) with error 5×10^{-7} have :

$$R_n \approx \int_n^{\infty} f(x)dx = \int_n^{\infty} x^{-5}dx = \left[\frac{x^{-4}}{-4}\right]_n^{\infty} = \frac{1}{4n^4},$$

$$\frac{1}{4n^4} \leq 5 \times 10^{-7} \Longrightarrow n^4 \geq \frac{1}{20} \times 10^7 \Longrightarrow n \geq \sqrt[4]{500{,}000} \approx 26.6\ ,$$

For (b) with error 5×10^{-7} have :

$$R_n \approx \int_n^{\infty} f(x)dx = \int_n^{\infty} 4x^{-7}dx = 4\left[\frac{x^{-6}}{-6}\right]_n^{\infty} = \frac{4}{6n^6},$$

$$\frac{4}{6n^6} \leq 5 \times 10^{-7} \Longrightarrow n^6 \geq \frac{4}{30} \times 10^7 \Longrightarrow n \geq \sqrt[4]{1{,}333{,}333} \approx 10.5 \ ,$$